

8. Would the coupling of the processes shown in Figure 17.24 be found in a eukaryotic cell? Explain why or why not.

9. Complete the following table:

Type of RNA	Functions
Messenger RNA (mRNA)	
Transfer RNA (tRNA)	
	In a ribosome, plays a structural role; as a ribozyme, plays a catalytic role (catalyzes peptide bond formation)
Primary transcript	
Small RNAs in the spliceosome	

Levels 5-6: Evaluating/Creating

10. **EVOLUTION CONNECTION** Most amino acids are coded for by a set of similar codons (see Figure 17.6). Propose at least one evolutionary explanation to account for this pattern.
11. **SCIENTIFIC INQUIRY** Knowing that the genetic code is almost universal, a scientist uses molecular biological methods to insert the human β -globin gene (shown in Figure 17.12) into bacterial cells, hoping the cells will express it and synthesize functional β -globin protein. Instead, the protein produced is nonfunctional and is found to contain many fewer amino acids than does β -globin made by a eukaryotic cell. Explain why.
12. **WRITE ABOUT A THEME: INFORMATION** Evolution accounts for the unity and diversity of life, and the continuity of life is based on heritable information in the form of DNA. In a short essay (100–150 words), discuss how the fidelity with

which DNA is inherited is related to the processes of evolution. (Review the discussion of proofreading and DNA repair in Concept 16.2.)

13. SYNTHESIZE YOUR KNOWLEDGE



Some mutations result in proteins that function well at one temperature but are nonfunctional at a different (usually higher) temperature. Siamese cats have such a “temperature-sensitive” mutation in a gene encoding an enzyme that makes dark pigment in the fur. The mutation results in the breed’s distinctive point markings and lighter body color (see the photo). Using this information and what you learned in the chapter, explain the pattern of the cat’s fur pigmentation.

For selected answers, see Appendix A.

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How is green fluorescent protein (GFP) used as a marker for gene expression? Go to “Lighting up Life” at www.scienceintheclassroom.org.

➔ **Instructors:** Questions can be assigned in Mastering Biology.